

## Comparing English Implicit Comparison and Japanese *Motto*-Sentences

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## Background (II): Explicit vs. implicit comparison

- Both the positive and comparative form of a gradable adjective can be used to express comparison (see the discussion on implicit comparison by Kennedy 2007, Sawada 2009):
- (2) a. Compared to Taro, Hanako is taller. **Explicit comparison: taller**  
(= Hanako is taller than Taro (is).)
- b. Compared to Taro, Hanako is tall. **Implicit comparison: tall**

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## Background (I): *tall* vs. *taller*

- Languages like English have a morphological distinction between the **positive** and **comparative** use of a gradable adjective: e.g., *tall* vs. *taller*.
- (1) The positive use vs. the comparative use of English *tall*
- a. (i) Hanako is very tall. **Positive use: ✓ tall**  
(ii) \*Hanako is very taller. **Positive use: # taller**
- b. (i) \*Hanako is tall than Taro (is). **Comparative use: # tall**  
(ii) Hanako is taller than Taro (is). **Comparative use: ✓ taller**

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## Background (III): Japanese gradable adjectives

- Languages like Japanese do not have a comparative morpheme like English *-er*, and thus, there is no morphological distinction between the positive and comparative form of a gradable adjective:
- (3) Hanako-wa se-ga takai.  
Hanako-TOP back-NOM tall  
'Hanako is tall.' **Positive use: takai**
- (4) Hanako-wa Taro-yori se-ga takai.  
Hanako-TOP Taro-STDD back-NOM taller  
'Hanako is taller than Taro.' **Comparative use: takai**

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## Background (IV): Japanese *motto*-sentences

- In Japanese, a morpheme like *motto* can be optionally included.
- (5) Hanako-wa Taro-yori motto se-ga takai.  
Hanako-TOP Taro-STDD MOTTO back-NOM tall(er)  
It seems to mean ‘Hanako is even taller than Taro (is).’
- What is the meaning contribution of *motto*?
  - Does a Japanese *motto*-sentence involve an implicit comparison or an explicit comparison?
  - How do we interpret a *motto*-sentence?

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## Today’s take-home messages

- (5) Hanako-wa Taro-yori motto se-ga takai.  
Hanako-TOP Taro-STDD MOTTO back-NOM tall(er)  
It seems to mean ‘Hanako is even taller than Taro (is).’  
‘Compared to Taro, Hanako’s height reaches a new level.’
- *Motto*-sentences express **implicit comparison** (not explicit comparison), involving a positive use (not comparative use) of gradable adjectives.
  - *Motto* semantically introduces **a new positive threshold between the items under comparison**: here Hanako’s height vs. Taro’s height.
  - For a *motto*-sentence, **the evaluative inference** that the standard already reaches the regular positive threshold (here ‘Taro is already tall’) is an **implicature**, not entailed meaning.
    - The contribution of *motto* is **different from even, but similar to new**:
- (6) Cato’s reputation, far from being diminished by Caesar’s attack, was raised to new heights. (Tom Holland, 2003, *Rubicon*)  
~> Before Caesar’s attack, Cato’s reputation was already high.  
(implicature)

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## Outline

- 1 Empirical observations
- 2 The semantics of *motto*
- 3 The pragmatics of implicit comparison and *motto*-sentences
- 4 Concluding remarks

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## English implicit comparison vs. Japanese *motto*-sentences

- In English, there is a **morphological distinction** between implicit vs. explicit comparison:
- (2) a. Compared to Taro, Hanako is taller. **Explicit comparison: taller**  
(= Hanako is taller than Taro (is).)  
b. Compared to Taro, Hanako is tall. **Implicit comparison: tall**
- In Japanese, there is **no morphological evidence** to show the distinction between a positive and a comparative use.
  - Thus below we show that *motto*-sentences pattern with English implicit comparison semantically and/or pragmatically.

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## 1. Incompatibility with a numerical differential

- Both English implicit comparison and *motto*-sentences are never compatible with a numerical differential (see also [Beck et al. 2004](#)'s example (130)).
- (7) a. Compared to Taro, Hanako is 5 cm taller. Explicit comparison  
b. \*Compared to Taro, Hanako is 5 cm tall. Implicit comparison
- (8) a. Hanako-wa Taro-yori go senchi se-ga takai  
Hanako-TOP Taro-STDD 5 cm back-NOM tall  
'Hanako is 5 cm taller than Taro.'  
b. \* Hanako-wa Taro-yori motto go senchi se-ga takai  
Hanako-TOP Taro-STDD MOTTO 5 cm back-NOM tall  
Intended: 'Hanako is 5 cm taller than Taro.'

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## 2. Infelicity under a 'crisp judgment' context

- (9) 'Crisp judgment' context:  
a. Taro is 165 cm tall, while Hanako is 166 cm tall.  
b. Taro is 180 cm tall, while Hanako is 181 cm tall. Both are tall.
- (11) a. Hanako-wa Taro-yori se-ga takai  
Hanako-TOP Taro-STDD back-NOM tall  
'Hanako is taller than Taro.' Without *motto*: felicitous under (9)  
b. ?? Hanako-wa Taro-yori motto se-ga takai  
Hanako-TOP Taro-STDD MOTTO back-NOM tall  
With *motto*: infelicitous under (9)
- The *motto*-sentence (11b) cannot have the same meaning as (12) – 'even + explicit comparison' (cf. [Beck et al. 2004](#), [Sawada 2014](#), see also [Liu 2010](#), [Chen 2023](#), [Chen and Greenberg 2024](#) on similar phenomena).
- (12) a. Even compared to Taro, Hanako is taller. Felicitous under (9b)  
b. Hanako is even taller than Taro (is). Felicitous under (9b)

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## 2. Infelicity under a 'crisp judgment' context

- Implicit comparison and *motto*-sentences are infelicitous under a 'crisp judgment' context, where the difference between items under comparison is very slight (see [Kennedy 2007](#)).
- (9) 'Crisp judgment' context:  
a. Taro is 165 cm tall, while Hanako is 166 cm tall.  
b. Taro is 180 cm tall, while Hanako is 181 cm tall. Both are tall.
- (10) a. Compared to Taro, Hanako is taller.  
Explicit comparison is felicitous under (9)  
b. ??Compared to Taro, Hanako is tall.  
Implicit comparison is infelicitous under (9)

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## 3. Evaluative inferences

- Implicit comparison, but not explicit comparison, has evaluative inferences, which are implicatures (see [Sawada 2009](#)).
- (13) Compared to Taro, Hanako is tall. Implicit comparison  
a. (13) implies Taro is not tall.  
b. (13) implies Hanako is not definitely tall.
- Even* cancels the above inferences and brings different inferences:
- (14) Even compared to Taro<sub>F</sub>, Hanako is tall. *Even* + Implicit comparison  
a. (14) presupposes Taro is tall. (see [Greenberg 2018](#), [Zhang 2022](#))  
b. (14) entails Hanako is definitely tall.
- Evaluative inferences of (14) cannot be canceled or suspended.
- (15) Even compared to Taro, Hanako is tall. #But actually they are both short.
- (16) I don't know if Taro is tall. But (\*even) compared to Taro, Hanako is tall.

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### 3. Evaluative inferences

- *Motto*-sentences pattern with English implicit comparison (but not English explicit comparison) in having evaluative inferences.
- Apparently, *motto*-sentence (17) seems to share the same inferences as the *even*-containing implicit comparison sentence (14):

- (14) Even compared to Taro<sub>F</sub>, Hanako is tall. *Even* + Implicit comparison
- a. (14) presupposes Taro is tall.
  - b. (14) entails Hanako is definitely tall.

- (17) Hanako-wa Taro-yori motto se-ga takai  
Hanako-TOP Taro-STDD MOTTO back-NOM tall  
(17)  $\leadsto$  Taro is tall.  
(17)  $\leadsto$  Hanako is definitely tall.

- However, *motto*-sentence (17) is distinct from (14) in two aspects:
  - Stress is typically on *motto*, not on *Taro*, suggesting that *motto* does not work with an invoked alternative set as *even* does;
  - The evaluative inferences of (17) can be suspended or canceled.

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### 3. Evaluative inferences

- The optional use of *motto* in (19) is also felicitous under its context.
- Here Taro's complaint implicitly cancels the evaluative inference that his part is already large. In fact, Taro's complaint means that his part is not large.

- (19) Context: Hanako divides a cake into two parts to distribute between her and Taro. Taro complains that the part Hanako takes is significantly larger.

'fukouhei da yo. Kimi-no bun-no hou-ga (motto) ooi yo.'  
unfair DECL SFP your part-GEN side-NOM MOTTO big SFT

'It's unfair. Your part is significantly larger.'

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### 3. Evaluative inferences

- In (18), the evaluative inference that Taro is tall is suspended.
- The optional use of *motto* in (18) is felicitous as far as the speaker knows that the difference between Taro and Hanako is significantly large.

- (18) Taro-ga takai ka dou ka-wa wakara-nai kedo, Hanako-wa  
Taro-NOM tall Q how Q-TOP know-NEG CONJ., Hanako-TOP  
zettai-ni Taro-yori (motto) takai.  
definitely Taro-STDD (MOTTO) tall  
'Though I don't know whether Taro is tall or not, compared to Taro,  
Hanako is definitely tall.'

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### 3. Evaluative inferences

- This kind of negative evaluative inference with regard to comparison standard has also been studied by Sawada (2014).
- For Japanese *motto*-sentence with the negative implicature, (20), stress is on the predicate part.

- (20) kono mise-no keeki-wa motto oishi-katta  
this store-GEN cake-TOP MOTTO delicious-PST  
'This store's cake used to be delicious.'  $\leadsto$  It's not delicious now.  
(Sawada 2014)

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### 3. Evaluative inferences

- The optional use of *motto* is also felicitous in negative sentences in (21).
- Here the evaluative inference that the comparison standard reaches the positive threshold (i.e., Hanako did well) is explicitly canceled.

- (21) Hanako-wa umaku deki-na-katta kedo, dare-mo (motto) umaku  
 Hanako-TOP well can-NEG-PST CONJ. ANYONE MOTTO well  
 deki-na-katta.  
 can-NEG-PST  
 ‘Hanako didn’t do well, but no one did better than her.’

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### Empirical generalizations

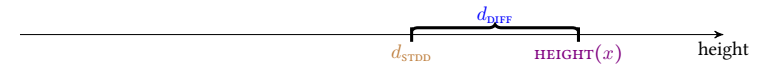
|                          | <i>Motto</i> -sentence | Implicit comparison | Explicit comparison |
|--------------------------|------------------------|---------------------|---------------------|
| Numerical differential   | incompatible           | incompatible        | compatible          |
| ‘Crisp judgment’ context | incompatabile          | incompatible        | compatible          |
| Evaluative inferences    | yes                    | yes                 | no                  |

- Japanese *motto*-sentences express implicit comparison.

- (4) Hanako-wa Taro-yori se-ga takai.  
 Hanako-TOP Taro-STDD back-NOM taller  
 Without *motto*: explicit comparison  
 ‘Compared to Taro, Hanako is taller.’ = ‘Hanako is taller than Taro.’
- (5) Hanako-wa Taro-yori motto se-ga takai.  
 Hanako-TOP Taro-STDD MOTTO back-NOM tall(er)  
 It seems to mean ‘Hanako is even taller than Taro (is).’  
 With *motto*: implicit comparison  
 ‘Compared to Taro, Hanako’s height reaches a new level.’

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### The semantics of gradable adjectives



- $\llbracket \text{tall} \rrbracket / \llbracket \text{takai} \rrbracket$  expresses a relation: there is a non-negative / positive distance,  $d_{\text{DIFF}}$ , between the measurement of the target,  $\text{HEIGHT}(x)$ , and the reference position,  $d_{\text{STDD}}$  (Zhang and Zhang-Yukun 2024).

$$(22) \quad \llbracket \text{tall} \rrbracket \stackrel{\text{def}}{=} \lambda d_{\text{DIFF}}. \lambda d_{\text{STDD}}. \lambda x. \lambda C. \underbrace{[d_{\text{DIFF}} \geq 0]}_{\text{non-negative presupposition}}. \{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(x) - d_{\text{STDD}}}_{x\text{'s height reaches reference position } d_{\text{STDD}}} \geq d_{\text{DIFF}} \}.$$

$$(23) \quad \llbracket \text{takai} \rrbracket \stackrel{\text{def}}{=} \lambda d_{\text{DIFF}}. \lambda d_{\text{STDD}}. \lambda x. \lambda C. \underbrace{[d_{\text{DIFF}} > 0]}_{\text{positive presupposition}}. \{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(x) - d_{\text{STDD}}}_{x\text{'s height exceeds reference position } d_{\text{STDD}}} \geq d_{\text{DIFF}} \}.$$

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## The positive use



[[Hanako is tall]] and [[Hanako-wa se-ga takai]]

- For the positive use, the reference,  $d_{\text{STDD}}$ , is the positive threshold.

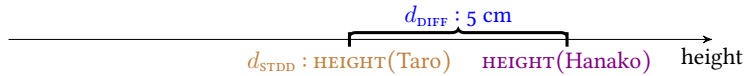
$$(24) \quad \llbracket \text{Hanako is pos}^{i_{\text{tall}}} \text{ tall} \rrbracket \Leftrightarrow \lambda C. \quad \underbrace{[d_{\text{DIFF}} \geq 0]}_{\text{non-negative presupposition}}.$$

$$\{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(\text{Hanako}) - d_{\text{tall-pos}}}_{\text{Hanako's height reaches the positive threshold } d_{\text{tall-pos}}} \geq d_{\text{DIFF}} \}.$$

$$(25) \quad \llbracket \text{Hanako-wa se-ga pos}^{i_{\text{tall}}} \text{ takai} \rrbracket \Leftrightarrow \lambda C. \quad \underbrace{[d_{\text{DIFF}} > 0]}_{\text{positive presupposition}}.$$

$$\{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(\text{Hanako}) - d_{\text{tall-pos}}}_{\text{Hanako's height exceeds the positive threshold } d_{\text{tall-pos}}} \geq d_{\text{DIFF}} \}.$$

## Explicit comparison with a numerical differential



[[Hanako is 5 cm taller than Taro]] and [[Hanako-wa Taro-yori go senchi se-ga takai]]

- A numerical differential (e.g., 5 cm) restricts  $d_{\text{DIFF}}$ .

$$(28) \quad \llbracket \text{Hanako is 5 cm taller than Taro} \rrbracket \Leftrightarrow \lambda C. \{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(\text{Hanako}) - \text{HEIGHT}(\text{Taro})}_{\text{Hanako's height exceeds Taro's height}} \geq d_{\text{DIFF}}, d_{\text{DIFF}} = 5 \text{ cm} \}.$$

$$(29) \quad \llbracket \text{Hanako-wa Taro-yori go senchi se-ga takai} \rrbracket \Leftrightarrow \lambda C. \{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(\text{Hanako}) - \text{HEIGHT}(\text{Taro})}_{\text{Hanako's height exceeds Taro's height}} \geq d_{\text{DIFF}}, d_{\text{DIFF}} = 5 \text{ cm} \}.$$

## Explicit comparison



[[Hanako is taller than Taro]] and [[Hanako-wa Taro-yori se-ga takai]]

- For the comparative use, the reference,  $d_{\text{STDD}}$ , is the measurement of the item corresponding to the target, here Taro's height.
- In English, comparative morpheme *-er* expresses an increase, restricting  $d_{\text{DIFF}}$  to a positive value, instead of a non-negative value (Zhang and Zhang-Yukun 2024).

$$(26) \quad \llbracket \text{Hanako is taller than Taro} \rrbracket \Leftrightarrow \lambda C. \{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(\text{Hanako}) - \text{HEIGHT}(\text{Taro})}_{\text{Hanako's height exceeds Taro's height}} \geq d_{\text{DIFF}}, d_{\text{DIFF}} > 0 \}.$$

$$(27) \quad \llbracket \text{Hanako-wa Taro-yori se-ga takai} \rrbracket \Leftrightarrow \lambda C. \quad \underbrace{[d_{\text{DIFF}} > 0]}_{\text{positive presupposition}}.$$

$$\{ \langle w, g \rangle \in C \mid \underbrace{\text{HEIGHT}_w(\text{Hanako}) - \text{HEIGHT}(\text{Taro})}_{\text{Hanako's height exceeds Taro's height}} \geq d_{\text{DIFF}} \}.$$

## Implicit comparison



[[Compared to Taro, Hanako is tall]]

$$(30) \quad \llbracket \text{Compared to Taro, Hanako is pos}^{i_{\text{tall}}} \text{ tall} \rrbracket \Leftrightarrow \lambda C. [d_{\text{DIFF}} \geq 0]. \{ \langle w, g \rangle \in C \mid \text{HEIGHT}_w(\text{Hanako}) - d_{\text{tall-pos}}^c \geq d_{\text{DIFF}}, d_{\text{tall-pos}}^c > \text{HEIGHT}(\text{Taro}) \}$$

- Implicit comparison is still a positive use of gradable adjectives, introducing a local-context-based positive threshold,  $d_{\text{tall-pos}}^c$ .
- $d_{\text{tall-pos}}^c$  has a restriction: it is between the measurement of the target and its corresponding item. Thus wrt their comparison with  $d_{\text{tall-pos}}^c$ , the two items under comparison yield qualitatively different results.
- The regular positive threshold,  $d_{\text{tall-pos}}$ , is not affected. Thus semantically, there is no positive inference for the target (here Hanako) or its corresponding item (here Taro).

## Japanese *motto*-sentence

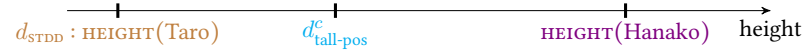


[[Hanako-wa Taro-yori motto se-ga takai]]

- (31)  $\llbracket \text{motto}^{i'_{\text{tall}}} \rrbracket \stackrel{\text{def}}{=} \lambda \mathcal{G}. \lambda d_{\text{CORR}}. \lambda x. \lambda C. \{ \langle w, g^{i'_{\text{tall}} \mapsto d_{\text{tall-pos}}^c} \rangle \mid \langle w, g \rangle \in \mathcal{G}(d_{\text{DIFF}})(d_{\text{tall-pos}}^c)(x)(C), d_{\text{tall-pos}}^c > d_{\text{CORR}} \}$
- (32)  $\llbracket \text{Hanako-wa Taro-yori motto}^{i'_{\text{tall}}} \text{ se-ga takai} \rrbracket \Leftrightarrow \lambda C. [d_{\text{DIFF}} \geq 0]. \{ \langle w, g^{i'_{\text{tall}} \mapsto d_{\text{tall-pos}}^c} \rangle \mid \langle w, g \rangle \in C, \text{HEIGHT}_w(\text{Hanako}) - d_{\text{tall-pos}}^c \geq d_{\text{DIFF}}, d_{\text{tall-pos}}^c > \text{HEIGHT}(\text{Taro}) \}$
- A *motto*-sentence also involves a positive use of gradable adjectives.
  - *Motto* introduces a local-context-based positive threshold,  $d_{\text{tall-pos}}^c$ , restricted to a value between the target and its corresponding item.

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## Implicit comparison vs. *motto*-sentence



[[Compared to Taro, Hanako is tall]] and [[Hanako-wa Taro-yori motto se-ga takai]]

- (30)  $\llbracket \text{Compared to Taro, Hanako is } \text{POS}^{i'_{\text{tall}}} \text{ tall} \rrbracket \Leftrightarrow \lambda C. [d_{\text{DIFF}} \geq 0]. \{ \langle w, g^{i'_{\text{tall}} \mapsto d_{\text{tall-pos}}^c} \rangle \mid \langle w, g \rangle \in C, \text{HEIGHT}_w(\text{Hanako}) - d_{\text{tall-pos}}^c \geq d_{\text{DIFF}}, d_{\text{tall-pos}}^c > \text{HEIGHT}(\text{Taro}) \}$
- (32)  $\llbracket \text{Hanako-wa Taro-yori motto}^{i'_{\text{tall}}} \text{ se-ga takai} \rrbracket \Leftrightarrow \lambda C. [d_{\text{DIFF}} \geq 0]. \{ \langle w, g^{i'_{\text{tall}} \mapsto d_{\text{tall-pos}}^c} \rangle \mid \langle w, g \rangle \in C, \text{HEIGHT}_w(\text{Hanako}) - d_{\text{tall-pos}}^c \geq d_{\text{DIFF}}, d_{\text{tall-pos}}^c > \text{HEIGHT}(\text{Taro}) \}$   
 ‘Compared to Taro, Hanako reaches a new height.’
- Implicit comparison (30) and *motto*-sentence (32) are semantically the same. However, the availability of a *motto*-sentence, in addition to a comparison sentence without *motto*, has pragmatic consequences.

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## Implicit comparison vs. *motto*-sentence

- Apparently, implicit comparison (13) and *motto*-sentence (17) do not have the same evaluative inferences.
- (13) Compared to Taro, Hanako is tall. Implicit comparison
- (13) implies Taro is not tall.
  - (13) implies Hanako is not definitely tall.
- (17) Hanako-wa Taro-yori motto se-ga takai  
 Hanako-TOP Taro-STDD MOTTO back-NOM tall  
 ‘Compared to Taro, Hanako’s height reaches a new level.’
- (17) implies Taro is tall.
  - (17) implies Hanako is definitely tall.
- Below we show how to derive their evaluative implicatures respectively under a simplified version of RSA (Rational Speech Act) modeling (see Lassiter and Goodman 2017, Degen 2023, among many others).
  - The differences between (13) and (17) hinge on their utterance space:
    - (13): the bare positive use, *motto*-less implicit comparison;
    - (17): the bare positive use, *motto*-less comparison, and a *motto*-sentence

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## Evaluative implicatures of implicit comparison

- **Meaning space:** 24 distinct ordering possibilities among
  - ▶ **Ha:** the height of Hanako
  - ▶ **Ta:** the height of Taro
  - ▶ **pos:** the regular positive threshold
  - ▶ **pos<sup>c</sup>:** the locally introduced positive threshold

Utterance space

Literal meaning

- $u_0$  = Hanako is tall.  $\text{Ha} \geq \text{pos}$
- $u_{ic}$  = Compared to Taro, Hanako is tall.  $\text{Ha} \geq \text{pos}^c > \text{Ta}$
- For a literal listener, among the 24 possibilities in the meaning space,
  - ▶ 13 possibilities are true under  $u_0$  or  $u_{ic}$  – we will only consider them
  - ▶ 12 possibilities are true under the positive use  $u_0$ ,  $\therefore$  each 0.08
  - ▶ 4 possibilities are true under the implicit comparison  $u_{ic}$ ,  $\therefore$  each 0.25

|          | $m_1$            | $m_2$            | $m_3$            | $m_4$            | $m_5$            | $m_6$            | $m_8$            | $m_{15}$ | $m_{16}$ | $m_{17}$         | $m_{19}$ | $m_{20}$         | $m_{24}$         |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|----------|----------|------------------|----------|------------------|------------------|
| $u_0$    | Ha               | Ha               | Ha               | Ha               | Ha               | Ha               | pos              | pos      | pos      | pos              | pos      | Ta               | Ta               |
| $u_{ic}$ | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ha               | Ha       | Ha       | pos              | pos      | pos <sup>c</sup> | pos <sup>c</sup> |
|          | Ta               | pos <sup>c</sup> | Ta               | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta       | Ha       | pos <sup>c</sup> | Ha       | pos              | Ha               |
|          | pos <sup>c</sup> | Ta               | pos              | Ta               | pos <sup>c</sup> | pos              | Ta               | pos      | Ta       | pos              | Ha       | pos <sup>c</sup> | pos              |
| $u_0$    | .08              | .08              | .08              | .08              | .08              | .08              | 0                | 0        | 0        | 0                | 0        | 0                | .08              |
| $u_{ic}$ | 0                | .25              | .25              | .25              | 0                | 0                | .25              | 0        | 0        | 0                | 0        | 0                | 0                |

Probability distribution for a literal listener  $L_0$

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## Evaluative implicatures of implicit comparison

Utterance space

Literal meaning

$u_0$  = Hanako is tall.

$\text{Ha} \geq \text{pos}$

$u_{ic}$  = Compared to Taro, Hanako is tall.

$\text{Ha} \geq \text{pos}^c > \text{Ta}$

|          | $m_1$            | $m_2$            | $m_3$            | $m_4$            | $m_5$            | $m_6$            | $m_8$            | $m_{15}$         | $m_{16}$         | $m_{17}$         | $m_{19}$         | $m_{20}$         | $m_{24}$         |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| $u_0$    | Ha               | Ha               | Ha               | Ha               | Ha               | Ha               | pos              | pos <sup>c</sup> | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ta               |
| $u_{ic}$ | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ha               | Ha               | Ha               | Ta               | Ha               | Ha               | pos <sup>c</sup> |
|          | Ta               | pos <sup>c</sup> | Ta               | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | pos              | Ha               | pos <sup>c</sup> | pos              | Ha               |
|          | pos <sup>c</sup> | Ta               | pos              | Ta               | pos <sup>c</sup> | pos              | Ta               | pos              | Ta               | pos              | pos              | pos <sup>c</sup> | pos              |
| $u_0$    | .08              | .08              | .08              | .08              | .08              | .08              | 0                | .08              | .08              | .08              | .08              | .08              | .08              |
| $u_{ic}$ | 0                | .25              | .25              | .25              | 0                | 0                | .25              | 0                | 0                | 0                | 0                | 0                | 0                |

Probability distribution for a literal listener  $L_0$

|          |   |     |     |     |   |   |   |   |   |   |   |   |   |
|----------|---|-----|-----|-----|---|---|---|---|---|---|---|---|---|
| $u_0$    | 1 | .25 | .25 | .25 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| $u_{ic}$ | 0 | .75 | .75 | .75 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |

Given a certain situation  $m$ , for a pragmatic speaker  $S_1$ , how useful an utterance is in identifying  $m$

|          |      |       |       |       |      |      |     |      |      |      |      |      |      |
|----------|------|-------|-------|-------|------|------|-----|------|------|------|------|------|------|
| $u_0$    | .103 | .0256 | .0256 | .0256 | .103 | .103 | 0   | .103 | .103 | .103 | .103 | .103 | .103 |
| $u_{ic}$ | 0    | .23   | .23   | .23   | 0    | 0    | .31 | 0    | 0    | 0    | 0    | 0    | 0    |

Probability distribution for a pragmatic listener  $L_1$

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## Evaluative implicatures of implicit comparison

Utterance space

Literal meaning

$u_0$  = Hanako is tall.

$\text{Ha} \geq \text{pos}$

$u_{ic}$  = Compared to Taro, Hanako is tall.

$\text{Ha} \geq \text{pos}^c > \text{Ta}$

|          | $m_1$            | $m_2$            | $m_3$            | $m_4$            | $m_5$            | $m_6$            | $m_8$            | $m_{15}$         | $m_{16}$         | $m_{17}$         | $m_{19}$         | $m_{20}$         | $m_{24}$         |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| $u_0$    | Ha               | Ha               | Ha               | Ha               | Ha               | Ha               | pos              | pos <sup>c</sup> | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ta               |
| $u_{ic}$ | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ha               | Ha               | Ha               | Ta               | Ha               | Ha               | pos <sup>c</sup> |
|          | Ta               | pos <sup>c</sup> | Ta               | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | pos              | Ha               | pos <sup>c</sup> | pos              | Ha               |
|          | pos <sup>c</sup> | Ta               | pos              | Ta               | pos <sup>c</sup> | pos              | Ta               | pos              | Ta               | pos              | Ha               | pos <sup>c</sup> | pos              |
| $u_0$    | .103             | .0256            | .0256            | .0256            | .103             | .103             | 0                | .103             | .103             | .103             | .103             | .103             | .103             |
| $u_{ic}$ | 0                | .23              | .23              | .23              | 0                | 0                | .31              | 0                | 0                | 0                | 0                | 0                | 0                |

Probability distribution for a pragmatic listener  $L_1$

- For a pragmatic listener, given  $u_{ic}$ , ‘Compared to Taro, Hanako is tall’,
  - ▶ The ordering  $m_8$  ‘pos >  $\text{Ha} \geq \text{pos}^c > \text{Ta}$ ’ has the highest probability.
  - ▶ The probability of ‘pos > Ta (Taro is below the regular threshold)’ is 0.77.
- We haven’t considered **cost** in estimating the utility of the pragmatic speaker  $S_1$ .
  - ▶ If we consider cost, the utility of  $u_{ic}$  under  $m_2$ ,  $m_3$ , and  $m_4$  would be further lower.
  - ▶ Consequently, given  $u_{ic}$ , the probability of  $m_2$ ,  $m_3$ , and  $m_4$  for  $L_1$  would be further lower, and the probability of  $m_8$  will be further higher.

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## Evaluative implicatures of a motto-sentence

- **Meaning space:** 24 ordering possibilities among, **Ha**, **Ta**, **pos**, **pos<sup>c</sup>**

Utterance space

Literal meaning

- $u_0$  = Hanako is tall.

$\text{Ha} \geq \text{pos}$

- $u_{ic}$  = Compared to Taro, Hanako is tall.

$\text{Ha} \geq \text{pos}^c > \text{Ta}$

- $u_{motto}$  = Compared to Taro, Hanako’s height reaches a new level.

$\text{Ha} \geq \text{pos}^c > \text{Ta}$

- We assume
  - ▶ Adding a motto-sentence extends the previous utterance space.
  - ▶  $\text{cost}(u_{motto}) > \text{cost}(u_{ic}) > \text{cost}(u_0)$
  - ▶ The listener who encounters  $u_{motto}$  is already a pragmatic listener in interpreting  $u_0$  and  $u_{ic}$ .

|             | $m_1$            | $m_2$            | $m_3$            | $m_4$            | $m_5$            | $m_6$            | $m_8$            | $m_{15}$         | $m_{16}$         | $m_{17}$         | $m_{19}$         | $m_{20}$         | $m_{24}$         |
|-------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| $u_0$       | Ha               | Ha               | Ha               | Ha               | Ha               | Ha               | pos              | pos <sup>c</sup> | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ta               |
| $u_{ic}$    | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ha               | Ha               | Ha               | Ta               | Ha               | Ha               | pos <sup>c</sup> |
| $u_{motto}$ | Ta               | pos <sup>c</sup> | Ta               | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | pos              | Ha               | pos <sup>c</sup> | pos              | Ha               |
|             | pos <sup>c</sup> | Ta               | pos              | Ta               | pos <sup>c</sup> | pos              | Ta               | pos              | Ta               | pos              | pos              | pos <sup>c</sup> | pos              |
| $u_0$       | .103             | .0256            | .0256            | .0256            | .103             | .103             | 0                | .103             | .103             | .103             | .103             | .103             | .103             |
| $u_{ic}$    | 0                | .23              | .23              | .23              | 0                | 0                | .31              | 0                | 0                | 0                | 0                | 0                | 0                |
| $u_{motto}$ | 0                | .25              | .25              | .25              | 0                | 0                | .25              | 0                | 0                | 0                | 0                | 0                | 0                |

Probability distribution for a pragmatic listener  $L_1$  that encounters  $u_{motto}$

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## Evaluative implicatures of a *motto*-sentence

| Utterance space                                                      |                  |                  |                  |                  |                  |                  |                  | Literal meaning                           | Cost                              |                  |                  |                  |                  |  |
|----------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|-------------------------------------------|-----------------------------------|------------------|------------------|------------------|------------------|--|
| $u_0$ = Hanako is tall.                                              |                  |                  |                  |                  |                  |                  |                  | $\text{Ha} \geq \text{pos}$               |                                   |                  |                  |                  |                  |  |
| $u_{ic}$ = Compared to Taro, Hanako is tall.                         |                  |                  |                  |                  |                  |                  |                  | $\text{Ha} \geq \text{pos}^c > \text{Ta}$ | higher than $\text{Cost}(u_0)$    |                  |                  |                  |                  |  |
| $u_{motto}$ = Compared to Taro, Hanako's height reaches a new level. |                  |                  |                  |                  |                  |                  |                  | $\text{Ha} \geq \text{pos}^c > \text{Ta}$ | higher than $\text{Cost}(u_{ic})$ |                  |                  |                  |                  |  |
|                                                                      | $m_1$            | $m_2$            | $m_3$            | $m_4$            | $m_5$            | $m_6$            | $m_8$            | $m_{15}$                                  | $m_{16}$                          | $m_{17}$         | $m_{19}$         | $m_{20}$         | $m_{24}$         |  |
|                                                                      | Ha               | Ha               | Ha               | Ha               | Ha               | Ha               | pos              | pos <sup>c</sup>                          | pos <sup>c</sup>                  | pos <sup>c</sup> | Ta               | Ta               | Ta               |  |
|                                                                      | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta               | Ta               | Ha               | Ha                                        | Ha                                | Ta               | Ha               | Ha               | pos <sup>c</sup> |  |
|                                                                      | Ta               | pos <sup>c</sup> | Ta               | pos              | pos              | pos <sup>c</sup> | pos <sup>c</sup> | Ta                                        | pos                               | Ha               | pos <sup>c</sup> | pos              | Ha               |  |
|                                                                      | pos <sup>c</sup> | Ta               | pos              | Ta               | pos <sup>c</sup> | pos              | Ta               | pos                                       | Ta                                | pos              | pos              | pos <sup>c</sup> | pos              |  |
| $u_0$                                                                | .103             | .0256            | .0256            | .0256            | .103             | .103             | 0                | .103                                      | .103                              | .103             | .103             | .103             | .103             |  |
| $u_{ic}$                                                             | 0                | .23              | .23              | .23              | 0                | 0                | .31              | 0                                         | 0                                 | 0                | 0                | 0                | 0                |  |
| $u_{motto}$                                                          | 0                | .25              | .25              | .25              | 0                | 0                | .25              | 0                                         | 0                                 | 0                | 0                | 0                | 0                |  |

Probability distribution for a pragmatic listener  $L_1$  that encounters  $u_{motto}$

|             |   |        |   |        |   |   |       |  |   |   |   |   |   |   |
|-------------|---|--------|---|--------|---|---|-------|--|---|---|---|---|---|---|
| $u_0$       | 1 | high   | ? | high   | 1 | 1 | 0     |  | 1 | 1 | 1 | 1 | 1 | 1 |
| $u_{ic}$    | 0 | medium | ? | medium | 0 | 0 | > .55 |  | 0 | 0 | 0 | 0 | 0 | 0 |
| $u_{motto}$ | 0 | low    | ? | low    | 0 | 0 | < .45 |  | 0 | 0 | 0 | 0 | 0 | 0 |

Given a certain case, the utility of an utterance for a pragmatic speaker  $S_2$

- Given the high cost of  $u_{motto}$ , a pragmatic speaker  $S_2$  should estimate
  - Under  $m_8$ ,  $u_{ic}$  is the most useful utterance to identify  $m_8$ .
  - Under  $m_2$  or  $m_4$ ,  $u_0$  has the highest utility.
- Thus, based on  $S_2$ , a pragmatic listener  $L_2$  would assign the highest probability of  $u_{motto}$  to  $m_3$ :  $\text{Ha} \geq \text{pos}^c > \text{Ta} \geq \text{pos}$

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## Implicit comparison vs. *motto*-sentence

| Utterance space                                                             |                |                |                |                |                |                |                | Literal meaning                           |                | Cost                              |                |                |                |  |                |
|-----------------------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-------------------------------------------|----------------|-----------------------------------|----------------|----------------|----------------|--|----------------|
| $u_0$ = Hanako is tall.                                                     |                |                |                |                |                |                |                | $\text{Ha} \geq \text{pos}$               |                |                                   |                |                |                |  |                |
| $u_{ic}$ = Compared to Taro, Hanako is tall.                                |                |                |                |                |                |                |                | $\text{Ha} \geq \text{pos}^c > \text{Ta}$ |                | higher than $\text{Cost}(u_0)$    |                |                |                |  |                |
| $u_{\text{motto}}$ = Compared to Taro, Hanako's height reaches a new level. |                |                |                |                |                |                |                | $\text{Ha} \geq \text{pos}^c > \text{Ta}$ |                | higher than $\text{Cost}(u_{ic})$ |                |                |                |  |                |
|                                                                             | $m_1$          | $m_2$          | $m_3$          | $m_4$          | $m_5$          | $m_6$          | $m_8$          |                                           | $m_{15}$       | $m_{16}$                          | $m_{17}$       | $m_{19}$       | $m_{20}$       |  | $m_{24}$       |
|                                                                             | Ha             | Ha             | Ha             | Ha             | Ha             | Ha             | pos            |                                           | $\text{pos}^c$ | $\text{pos}^c$                    | $\text{pos}^c$ | Ta             | Ta             |  | Ta             |
|                                                                             | pos            | pos            | $\text{pos}^c$ | $\text{pos}^c$ | Ta             | Ta             | Ha             |                                           | Ha             | Ha                                | Ta             | Ha             | Ha             |  | $\text{pos}^c$ |
|                                                                             | Ta             | $\text{pos}^c$ | Ta             | pos            | pos            | $\text{pos}^c$ | $\text{pos}^c$ |                                           | Ta             | pos                               | Ha             | $\text{pos}^c$ | pos            |  | Ha             |
|                                                                             | $\text{pos}^c$ | Ta             | pos            | Ta             | $\text{pos}^c$ | pos            | Ta             |                                           | pos            | Ta                                | pos            | pos            | $\text{pos}^c$ |  | pos            |
| $u_0$                                                                       | .103           | .0256          | .0256          | .0256          | .103           | .103           | 0              |                                           | .103           | .103                              | .103           | .103           | .103           |  | .103           |
| $u_{ic}$                                                                    | 0              | .23            | .23            | .23            | 0              | 0              | .31            |                                           | 0              | 0                                 | 0              | 0              | 0              |  | 0              |
| $u_{\text{motto}}$                                                          | 0              | .25            | .25            | .25            | 0              | 0              | .25            |                                           | 0              | 0                                 | 0              | 0              | 0              |  | 0              |

Probability distribution for a pragmatic listener  $L_1$  that encounters  $u_{motto}$

|             |   |        |   |        |   |   |       |  |   |   |   |   |   |   |
|-------------|---|--------|---|--------|---|---|-------|--|---|---|---|---|---|---|
| $u_0$       | 1 | high   | ? | high   | 1 | 1 | 0     |  | 1 | 1 | 1 | 1 | 1 | 1 |
| $u_{ic}$    | 0 | medium | ? | medium | 0 | 0 | > .55 |  | 0 | 0 | 0 | 0 | 0 | 0 |
| $u_{motto}$ | 0 | low    | ? | low    | 0 | 0 | < .45 |  | 0 | 0 | 0 | 0 | 0 | 0 |

Given a certain case, the utility of an utterance for a pragmatic speaker  $S_2$

- Although  $u_{ic}$  and  $u_{motto}$  have the same semantics, they have different evaluative implicatures. For a pragmatic listener,
  - $u_{ic}$  implies  $\text{pos} > \text{Ha} \geq \text{pos}^c > \text{Ta}$
  - $u_{motto}$  implies  $\text{Ha} \geq \text{pos}^c > \text{Ta} \geq \text{pos}$
- We derive the implicature for  $u_{motto}$  based on the implicature of  $u_{ic}$ .

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## The use of *even* in implicit comparison or a *motto*-sentence

- (14) Even compared to Taro<sub>F</sub>, Hanako is tall. *Even* + Implicit comparison
- (14) presupposes Taro is tall.
  - (14) entails Hanako is definitely tall.
- (33) Hanako-no keeki-wa Taro-no keeki-yori-(mo) motto  
 Hanako-GEN cake-TOP Taro-GEN cake-STDD-(even) MOTTO  
 oishi-katta.  
 delicious-PST  
 Sawada (2014): 'Hanako's cake was even more delicious than Taro's.'
- (33) presupposes Taro's cake was already delicious.
  - (33) entails Hanako's cake was definitely delicious.
- In (14) and (33), the use of *even* / *mo* brings evaluative inferences that are entailed meanings.

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## Outline

- Empirical observations
- The semantics of *motto*
- The pragmatics of implicit comparison and *motto*-sentences
- Concluding remarks

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## Take-home messages

- (5) Hanako-wa Taro-yori motto se-ga takai.  
Hanako-TOP Taro-STDD MOTTO back-NOM tall(er)  
It seems to mean ‘Hanako is even taller than Taro (is).’  
‘Compared to Taro, Hanako’s height reaches a new level.’

- *Motto*-sentences express **implicit comparison** (not explicit comparison), involving a positive use (not comparative use) of gradable adjectives.
- *Motto* semantically introduces **a new positive threshold between the items under comparison**: here Hanako’s height vs. Taro’s height.
- For a *motto*-sentence, **the evaluative inference** that the standard already reaches the regular positive threshold (here ‘Taro is already tall’) is an **implicature**, not entailed meaning.

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# Thank you!

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